Hadi Lashkari Ghouchani

Software Engineer

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Summary

Software engineer with 21+ years of development experience, including 11 years of professional Android development and, most recently, mobile CI/CD and developer-experience engineering. Skilled in software architecture, algorithm design, and applied machine learning.
- Builds scalable, maintainable, and efficient systems, from low-level Android internals to app architecture
- Automates mobile delivery with trunk-based release pipelines, reproducible Nix builds, and AI-agent workflows
- Advocates SOLID principles, single source of truth, pure functions, and stateless design
- Applies machine learning to problems that make a real difference in people's lives
- Taught algorithms and programming at university level
- Continuous learner and open-source contributor
- Makes technical decisions proactively and transparently

Looking to join a creative, motivated team working on challenging technical problems.

Experience

Senior Mobile Developer

at Bell Canada | April, 2026 - Present | Toronto, Canada

Bell Canada is a Canadian telecommunications company

Mission:

- Took ownership of GitLab CI/CD and reversed a declining trend in pipeline job success rate, raising it from 18% to 31% (a 72% relative improvement) within six months
- Designed and implemented trunk-based development pipelines for Android and iOS across multiple repositories, automating release-branch creation, cherry-picks, publishing, and release-candidate promotion
- Built a multi-agent CI workflow that turns Jira tickets into merge requests, with planner, coder, and reviewer agents committing an initial implementation
- Automated DTO and mapper generation from APIDog API specs in CI, eliminating hand-written boilerplate from developer merge requests
- Restructured merge request pipelines into fast checks, required heavy jobs, and optional jobs, and strengthened automated code verification, shortening feedback loops and raising confidence before merge
- Made builds and emulator runs reproducible by wrapping them in Nix flakes
- Migrated GitLab jobs to custom, task-specific Linux containers, improving performance, isolation, and reproducibility
- Reduced GitLab SAST security findings to zero across multiple projects
- Pinned dependencies, tuned CI/CD caching, and established architecture and release-readiness practices
- Maintain and tune the EC2 instances running CI/CD for reliability and performance

Technologies:

- Android, Kotlin Multiplatform, Gradle Plugin
- iOS, Swift, XCode, XCodebuild
- CI/CD, Giltab's Configurations, Gitlab SAST, Podman, Nix, AWS EC2

Senior Mobile Developer

at Block Inc. - Bitkey project(via Touchlab) | November, 2025 - Feb, 2026 | Toronto, Canada

Block, Inc. is an American technology company and a financial services provider for consumers and merchants. The Bitkey is a project in Block Inc. that I directly work on. Bitkey is a digital wallet for bitcoin, available in the U.S. and 95 other countries.

Mission:

- Worked on multi backup project
- Fixed flaky tests, including unit and functional tests by solving multiple race conditions and state management problems
- Worked on the new onboarding flow

Technologies:

- Kotlin Multiplatform/KMP, Compose Multiplatform/CMP, Gradle Plugin
- iOS, Swift, XCode
- Rust binding, UniFFI
- Ktor, Anvil, SqlDelight, Kotest
- Github, Linear, Notion, BuildKite, etc.

Lead Android Developer

at CIBC (via Vision Talent) | March, 2025 - November 2025 | Toronto, Canada

The Canadian Imperial Bank of Commerce is a Canadian multinational banking and financial services corporation headquartered at CIBC Square in the Financial District of Toronto, Ontario.

Mission:

- Rewriting features in the newly developing CIBC mobile apps
- Improved the shared CI pipeline build-time by almost 50%, mostly by using build caches
- Initiated and Implemented memory management by applying structural DI scopes to release resources, ViewModels(in MVVM architecture), and their state
- Working on remote FeatureFlag functionality
- Migrated a Confluence's scraping script to use Selenium

Technologies:

- Kotlin Multiplatform/KMP, Android, Compose Multiplatform/CMP, Gradle Plugin, code generator
- iOS, Swift, XCode, SwiftUI
- Ktor, Koin, Compose Navigation, etc.
- Atlassian Jira, Atlassian Confluence, Github, Jenkins, etc.

Software Engineer -> Productivity Engineer

at Bell Canada (via MobileLIVE) | Aug, 2022 - Nov, 2024 | Toronto, Canada

Promoted to Productivity Engineer.

Mission:

- Resolved VPN network proxy and self-signed certificate issues by developing an automated installation script and Compose Multiplatform app, resulting in a 164% increase in build count on the Gradle Enterprise(Develocity) dashboard.
- Initiated and implemented Gradle Conventional Plugins to reduce code duplication and errors. Established Single Source of Truth (SSOT) for external library versions, significantly reducing runtime crashes caused by version conflicts.
- Spearheaded the adoption of Merge Trains and Merge Request Pipelines in GitLab, creating a Proof of Concept (POC) workflow that demonstrably improved code quality and streamlined the integration process.
- Optimized test execution, reducing incremental runtime by 52% and improving overall development efficiency.
- Served as the primary resource for build-related training and support across a team of 50+ Android developers, enhancing team productivity and knowledge sharing.
- Implemented a custom No-Framework dependency injection solution across multiple app and library repositories, enhancing modularity, testability, and maintainability while minimizing build times by avoiding code generation.
- Integrated Detekt to verify code quality in the merge request's pipelines.

Additional Technologies:

- Develocity(Gradle Enterprise), Gradle Conventional Plugin, Detekt
- Kotlin Multiplatform, Compose Multiplatform

Software Engineer

at Bell Canada (via MobileLIVE) | Apr. 2021 - Aug. 2022 | Toronto, Canada

MobileLIVE: A Canadian tech-service provider specializing in digital transformation solutions. Bell Canada: Canada's largest telecommunications company, offering mobile services (Bell Mobility, Virgin Mobile), television (Bell Satellite TV, Bell Fibe TV), and internet services across the country.

Mission:

- Implemented a Request for Comments (RFC) process for technical decision-making, streamlining the evaluation and adoption of new ideas across the organization.
- Led regular knowledge-sharing discussions across the team on SOLID principles, modularization best practices, feature flag implementation, and monorepo architecture, elevating the team's overall code quality and architectural design skills.
- Conceived and executed an automated system for cloning and building source code across a multi-repo setup, significantly reducing version conflicts and enhancing code visibility for efficient debugging.

Technologies:

- Kotlin Coroutines / Mockk, Mockito, jUnit, etc.
- Agile, Sprint, Lane based branching strategy, CI/CD
- Atlassian Jira, Atlassian Confluence, Gitlab, Jenkins, Slack, Branch.io, inVision, Dynatrace, etc.

Staff Android Developer

at iFood (via Letgo) | Oct. 2020 - Mar. 2021 | Barcelona, Spain

iFood is a leading food delivery platform, dominating markets in Brazil, Colombia, and Mexico. It's the market leader in Latin America with 50+ million Android app installations. Mission-critical achievement: Initiated and executed a strategy to refactor legacy code out of the root module, resulting in a 55% decrease in incremental build time and improved overall system architecture.

Senior Android Developer

at Letgo: Leading Classified Marketplace | Oct. 2019 - Mar. 2021 | Barcelona, Spain

One of the largest classified marketplaces in the United States at the time. Market leader in Turkey with 15+ million Android app installations. Mission-critical achievement: Optimized feed loading performance, reducing launch time by over 60 seconds.

Lead Android Developer

at Sheypoor: Top-Tier Classified Marketplace | Jan. 2018 - Jan. 2019 | Tehran, Iran

Impressive user base: 13+ million overall users, 1.5+ million daily active users. High engagement: 344+ million sessions. Key Achievement: Spearheaded initiatives that tripled daily active users from 500,000 to 1.5+ million within one year.

Senior Android Developer

at Myket: Top-Tier App Store and Social Network | Sep. 2015 - Des. 2017 | Tehran, Iran

Java and C# developer

at Soha System | Mar. 2015 - Aug. 2015 | Tehran, Iran

Public Contributions

- 2026 Tabular center: State machines whose declaration is the transition matrix
- 2026 Type Neural Network: A machine learning architecture based on the Type Mechanics
- 2026 MFFT benchmark: A benchmark among matmul implementations with focus on MFFT
- 2025 Post: Type Mechanics
- 2025 Post: Geometrical Probability
- 2024 Post: Matrix Fast Fourier transform(MFFT) - An optimization for Matrix Multiplication
- 2024 C Numbers: This is a collection of number types, including floating numbers and p-adic numbers.
- 2024 Post: Not All codes are liabilities
- 2024 City of Toronto RAG GPT: The project received the second position award in the TMLS Hackathon.
- 2024 dragon detector: A simple ML model to detect dragon in images
- 2024 Nix Effect pod: Modularize effects of all kinds, and stay in-control, with NixOS and podman.
- 2023 Contributing to nixpkgs/androidenv: Nix is a language that is designed to have reproducibile output.

Education

Sharif University Of Technology	M.Sc	in Physics	Feb. 2010	Tehran, Iran
Payame Noor University	B.Sc	in Physics	Jul. 2007	Mashhad, Iran

Tools / Skills

Kotlin Multiplatform/KMP

Compose Multiplatform/Android/iOS/Desktop
Advocate for independent UI/UX from platform
Main Contributor of Log4K
Main Contributor of LifecycleHandler

Android

MVVM/Android Architecture Components/JetPack
Anvil/Dagger 2/RxJava 2/Retrofit 2/...
Kotlin Multiplatform/Kotlin Coroutines
custom Kotlin Compiler Plugins/...
custom KSP plugins/...
Jetpack Compose/View Binding/...
Contributor of NixOS's Android Env

iOS

Swift/SwiftUI

Software Architecture

Clean Architecture/Hexagonal Architecture
Unidirectional data flow Architecture
Stateless Architecture
Logic Tree Architecture
MVVM/MVI/MVP/RMVVM

Principle

Clean Code/SOLID
Object Oriented Programming
Functional Programming
Isolation/Reproducibility
Stable Dependencies Principle
Stable Abstractions Principle

Design Pattern

Factory/Builder/Singleton/...
Adapter/Facade/Proxy/Visitor/Delegate/...
Strategy/Chain of responsibility/...

Modularization Strategies

Applying DIP/Decrease incremental build time
Code isolation to reduce dependencies

Continuous Integration/Deployment

Jenkins, GitLab CI, Fastlane

Android Performance Optimization

memory management/threading/profiling tools

Android Security

encryption/secure networking/permissions model

Android Test and Verification

Unit/Integration/Mockito/Espresso/JaCoCo/...
Detekt plugins/Konsist

Android Build System

ProGuard/R8/custom Gradle plugins

Build Tool

Nix/Gradle/Cargo/Sbt/Maven/Ant/Make/...

Command Line Tool

tmux/bash/zsh/git/grep/sed/netstat/awk/podman/...
adb/sdkmanager/avdmanager/xcodebuild/...

Rust

actix/rusqlite/tokio/thiserror/crossbeam
tonic/prost/openssl/flatbuffers/thrift
Rc/Arc/Mutex
FFI

Podman/Docker

Buildkit/Compose/Cache

Nix

NixOS/Nix Darwin/Home Manager
mkDrivation/mkShell

Programing Language

Kotlin/Rust/Python/Java/Scala
Groovy/Dart/Javascript/Ruby...

Workflow

Trunk Based development/Github flow/Git Flow
Cross-functional team

Operating System

Linux(NixOS/Fedora/Ubuntu/Debian), MacOS

Achievements and Recognition

- 2026 Had an offer from Bell Canada
- 2025 Had an offer from Block Inc. via Touchlab
- 2025 Had an offer from CIBC via Vision Talent
- 2024 Had an offer from Accenture via Experis
- 2024 First position in National Health Datathon
- 2024 Second position in AWS Summit's Gameday
- 2024 Second position in Toronto Machine Learning #TMLS Hackathon
- 2020 Had an offer from MobileLIVE, Toronto, Canada
- 2020 Had an offer from iFood, Remote
- 2019 Had an offer from Kik, Kitechener, Canada
- 2019 Had an offer from Letgo, Barcelona, Spain
- 2019 Had an offer from CRITICAL Techworks, BMW, Porto, Portugal
- 2019 Had an offer from The Next Big Thing, Berlin, Germany
- 2018 Received result-oriented badge in Sheypoor
- 2007 Ranked 44th among more than five thousand participants in the National Graduate Entrance Exam in Physics

Language

English: Advance
Farsi: Native

The latest version of this resume is available in github
Built on October 3, 2026